Bug Analysis Report: Event Booking Analytics Engine

1. Executive Summary

The initial implementation of the Event Booking Analytics Engine for *EventSphere* contained three critical bugs that corrupted revenue calculations, improperly mutated attendee data during filtering, and failed to aggregate popular event metrics correctly. This report documents these issues, their underlying root causes, and the engineering resolutions applied.

2. Bug Analysis Table

Bug Number	Problem	Root Cause	Resolution
Bug 1	Incorrect Revenue Calculation `(calculateRevenue)`	The revenue function only summed up the base `booking.ticketPrice` without taking into account how many tickets were purchased for each booking.	Updated the `.reduce()` logic to multiply tickets by price: `booking.tickets * booking.ticketPrice`.
Bug 2	Attendee Filtering Mutates Data `(getAttendees)`	The filter condition used an assignment operator (=) instead of a comparison operator (===).	Replaced = with === (booking.attended === true) to properly evaluate and filter attendee records without data mutation.
Bug 3	Popular Event Aggregation Fails (NaN) `(getPopularEvent)`	The aggregation logic attempted to increment counts on `eventMap[booking.eventName]` without checking if the property was initialized.	Initialized the map keys safely using logical OR fallback: `eventMap[booking.eventName] = (eventMap[booking.eventName]`

3. Summary of Code Improvements

- **Accurate Data Metrics:** Ensured mathematical calculations properly weigh multipliers like ticket quantities.
- **Pure Functions:** Removed unintended side effects and data mutations from filtering methods.
- **Safe Accumulation:** Implemented defensive initialization patterns for lookup maps during object aggregation.